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ORGANIZATIONAL CONTEXT AND SCIENTIFIC PRODUCTIVITY*

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An earlier study found that while a scientist's productivity does not affect the prestige of the academic position obtained, the prestige of the position does affect later productivity. In this paper consideration of contextual effects is broadened to include differing organizational contexts of scientific employment. Chances of obtaining employment in a particular context are not strongly affected by productivity. Once employment is obtained in a specific context, individual levels of productivity soon conform to characteristics of that context. These results do not support the idea that scientists are allocated to organizational contexts on the basis of their scientific contributions. Past research indicating that the most productive scientists are recruited to academic locations may have confused the cause of a scientist working in a given context with the effect of working in that context.

Studies of the scientific career are generally limited to scientists who hold faculty positions in Ph.D.-granting institutions, implicitly suggesting that scientists outside academia are not scientists in the same sense as those in academia. For example, Marcson (1960) and Kornhauser (1962) contrast professional scientists with organizational scientists; Hagstrom (1965) contrasts the "scientific community" with industrial research settings. The implicit assumption is that real science is performed only in the university. This assumption is based at least in part on a second assumption: the normative structure of science—as proposed by Merton (1973) and others (Hagstrom, 1965; Storer, 1966)—operates differentially among sectors of employment.

According to the normative structure of science, contributions to the body of scientific knowledge are the fundamental criteria of evaluation. The scientist is viewed as operating in a process of exchange—

knowledge for prestige and recognition (Hagstrom, 1965:13). The norm of communality requires that knowledge be made public for the gain of the community and not directly for that of the individual or employer. Such a model of science is based on science as performed in an academic setting (cf. Sklair, 1972), not in an industrial, developmental laboratory.

In nonacademic locations, particularly industry, the criteria are thought to be vastly different. Kornhauser (1962:13) claims that the requirements for success in an organization commonly clash with those for success in science as a profession. Industrial locations may control publications (although there are important exceptions), if not stop them altogether. The emphasis on the free exchange of information that is suggested by the functional theory of science is replaced by the more pragmatic logic of business. Publications must be delayed until their results are adequately covered by patents (cf. Cotgrove and Box, 1970:19).

Most studies of nonacademic scientists (Marcson, 1960; Kornhauser, 1962; Glaser, 1964) have focused on the difficulties that scientists may have in adapting to the perhaps deviant environment of industry: difficulties that arise from a conflict between the norms which scientists acquired during their graduate edu-

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cation and the expectations of their industrial employers. Cotgrove and Box (1970) indicate that this conflict is not necessarily as strong as previously argued, and may be almost nonexistent due to the differing motives of those who work in industry as opposed to academia. The suggestion of differing motives between nonacademic and academic scientists has also been noted by Krohn (1971:107), who states, "The argument has been that universities, government and industry respectively shape their research and people who do it. This socialization begins with the process of selective recruitment, whereby institutions and people reciprocally influence each other."

Regardless of the extent of conflict between nonacademic scientists and their employers, the choice of the organizational context of employment is an extremely important one. The context determines the research setting and facilities. It affects not only professional prestige, since academia is generally considered the most prestigious and desirable location and industry the least (Zuckerman, 1970), but also the opportunities for research in terms of freedom to publish and intellectual autonomy. Further, the choice of a nonacademic career is often final (Harmon, 1968:10); industrial employment almost assures that a return to academia will be blocked (Caplow and McGee, 1958:150).

Organizational context may be related to scientific productivity in a variety of ways. First, as Reskin (1977:493) has noted, the different goals of various locations may either hinder or promote productivity. For example, promotion in research universities is generally dependent upon publications. In industrial settings, publications are generally not criteria for promotion. The fact that such locations do not encourage publications, but rather tolerate them, may result in an increasingly negative effect of nonacademic locations on levels of publication through a process of reinforcement. Second, several studies (Caplow and McGee, 1958:150; Cotgrove and Box, 1970:2; Klaw, 1968:78; Marcson, 1960:54) have suggested that nonacademic locations have difficulties in recruiting the best graduates and must

settle for the second echelon. If so, the lesser abilities of nonacademic scientists would contribute to their lower productivity.

While numerous studies have contributed to our understanding of stratification within the academic sector (Allison, 1976; Cole and Cole, 1973; Gaston, 1978; Hagstrom, 1971; Hargens and Hagstrom, 1981; Long, 1978; Long et al., 1979), very few have emerged to explicate stratification in science as a whole, including both the academic and nonacademic sectors (Reskin, 1976, 1977, 1979). Yet to understand the entire social system of science, the significant and growing nonacademic sector must be considered. Further, to fully understand stratification within academia, the process leading to an academic career as opposed to a career in industry or government must be understood. To fill a gap in past research we provide a longitudinal analysis of productivity and position for a representative sample of both academic and nonacademic scientists. Three major questions are addressed. First, what determines the allocation of scientists to various organizational contexts? Specifically, do more productive scientists obtain positions in more prestigious locations? Second, to what extent does organizational context influence a scientist's productivity? Does a scientist begin to conform to the perhaps alien expectations of a nonacademic environment? Finally, are differences in productivity primarily the consequence of recruiting the more productive scientists in some sectors, or are they the result of working in different organizational contexts?

DATA AND MEASUREMENT

Analysis is based on the population of male biochemists who obtained their doctorates in the fiscal years 1957, 1958, 1962, and 1963. Female doctorates were excluded from consideration due to their small number and the difficulty encountered in obtaining complete biographic information on their careers. Biographic information was coded from the 10th through 13th editions of *American Men (and Women) of Science*. Complete in-

formation on educational and occupational experiences was obtained for 557 (83%) of the 668 males who obtained their degrees during this period. Prestige of the doctoral department was measured with the 3-digit ratings of faculty quality of biochemistry departments, a partial listing of which appeared in Cartter (1966). The prestige of postdoctoral appointments in graduate departments was based on a weighted average of the Roose and Andersen (1970) prestige scores for bioscience departments in the fellowship institution.¹ An average was used since the specific department of the postdoctoral appointment was not always known. The prestige scores range from a low of 1.00 to a high of 5.00. Whether a graduate department was located in a college of agriculture was also coded, a distinction in keeping with biochemistry's twin origins in medical and agricultural science (cf. Bernal, [1954] 1971:886ff.).

While the prestige of the doctoral department is often considered in studies of the scientific career, characteristics of a student's dissertation supervisor, hereafter referred to as the mentor, are generally ignored. Yet the mentor may be an important influence both in training and sponsoring the student. Using the American Chemical Society's *Directory of Graduate Research, Dissertation Abstracts*, and inquiries to specific universities, the name of the mentor was obtained for all but two of our sample. Following the work of Reskin (1979), two basic characteristics of the mentor were isolated: eminence and performance.

Performance reflects current (at the time of the student's doctorate) research productivity. While several measures of performance were collected, the indicator used is the number of citations (to both junior and senior authored papers) received by the mentor to papers published in the three-year period beginning with the student's doctorate. Citations were coded from the 1961 and 1966 volumes of *Science Citation Index*. Scientific eminence, while stemming from scientific performance, may lag performance and persist after performance declines (Zuckerman

and Merton, 1971:81 and 1972:325, as cited by Reskin, 1979:131). A number of indicators of eminence were coded, ranging from election to the National Academy of Science to the numbers of honorary degrees, postdoctoral fellowships, or scientific awards received. The measure used in the following analysis is a weighted count of prestigious awards such as the Nobel Prize and election to the National Academy of Science (weighted 4), honorary degrees (weighted 3), prestigious fellowships such as a Fulbright or Guggenheim (weighted 2), and less prestigious awards such as the Borden or Vaughan Awards (weighted 1). While more refined weightings based on prestige of the awards, such as those presented by Cole and Cole (1973:270-5) for physics, would be preferable, such a list is not available for biochemistry; and there were too many differences in the specific awards in biochemistry compared to physics to usefully apply the Coles' list. Details on the construction of this measure are presented in Long and McGinnis (1981).

Two measures of the employment experiences of the mentor were also coded. The contexts in which the mentor has worked may influence the context in which the student obtains employment. These measures are described below.

Productivity of the sample members was measured using counts of both publications and citations to them. *Chemical Abstracts* (1955-1973) was used to locate the articles published by the sample members, whether or not they were the senior authors. Citations to these articles were coded from *Science Citation Index* (Vols. 1961, 1964, 1966, 1968, 1970, 1972, and 1974). On multiple-authored papers where the cohort member was not the first author, the name of the first author was used to locate citations to junior authored papers (cf. Long et al., 1980 for details on this problem); thus downward bias in counts for scientists who were predominantly junior authors was avoided. For a given year in the scientist's career, the publication measure reflects publications in a three-year period ending in that year. The citation measure for that year is restricted to citations to papers published in that three-year period. Since coverage of *Science Citation Index* and *Chemical Ab-*

¹ The complete Roose and Andersen prestige scores were kindly provided by Charles J. Andersen.

tracts increased during the period covered by our analyses, counts were standardized within years of the Ph.D. For further details, see Long (1978).

CLASSIFICATION OF PRIMARY WORK ACTIVITY AND ORGANIZATIONAL SECTOR

A clear classification of organizational contexts is necessary to study the process by which scientists are allocated to various contexts and the effects of these contexts on scientific productivity. Two basic dimensions of context must be considered: employment sector and primary work activity.

The first dimension of organizational context is sector of employment. In his review of research settings, Marcson (1972) identifies the four sectors of government, industry, nonprofit, and university, with a secondary distinction between applied and basic research. Pelz and Andrews (1966) compare Ph.D.s in development-oriented laboratories (mainly industry and parts of government) with those in research-oriented laboratories (academia and parts of government). Harmon (1968) uses four divisions: academia, business and industry, government, and all other locations. Reskin (1977) classifies organizations in terms of their research orientations with two dummy variables: the first distinguishing university (but not college) locations from other locations; the second distinguishing research-oriented contexts from other contexts. Research-oriented contexts include universities, government laboratories, and nonprofit institutes. Unlike the other studies, Reskin notes the fundamental importance of the distinction between research universities and four-year colleges.

The second dimension concerns the work activities or roles which scientists perform. The three major activities distinguished by Harmon (1965) and later analyzed by Zuckerman and Merton (1972) are teaching, research, and administration. Most scientists spend some time in each of these roles, and accordingly one would prefer to know the percentages of time a scientist devotes to each of these activities. Unfortunately, such information is not accessible from secondary sources such as those used in this study.

Accordingly, classifications into what appeared to be modal work activities were made on the basis of the job titles given in *American Men and Women of Science*. Within industrial settings and research laboratories titles such as lab director, section head, and vice president for research were used to indicate administrative duties. Titles such as associate scientist, research scientist and biochemist were taken to indicate primarily research activities. Faculty positions, which often combine both teaching and research roles, were classified on the basis of job titles such as assistant, associate, or full professor.

The distinctions among research, teaching, and administration are not in themselves completely adequate. Clearly, teaching in a research university has a strong and perhaps dominant research component, while teaching in a four-year liberal arts college may have a negligible research component. Research carried out in a research university will be quite different from that done in most industrial laboratories. These distinctions should reflect the extent to which various organizational settings encourage the publication of research results. Within the academic sector, universities strongly encourage publication, if not outright demand it, while in four-year colleges and nonresearch universities it may be simply tolerated. Among research laboratories, locations such as academic research centers, private-nonprofit laboratories, and those exceptional industrial laboratories (e.g., Bell Laboratories and Sandia Laboratory of AT&T) encourage publication, while the majority of industrial laboratories and some federal laboratories restrict publication. Given the variety of research settings within some of the largest employers (e.g., General Electric), it is impossible to be certain if a scientist is in a location which encourages or discourages publication. Yet, by following the suggestions of past research, our division of employers into those who encourage publication and those who discourage publication should be quite accurate.

In our final classification, both primary work activity and sector are considered. The resulting classes are referred to as organizational contexts. Six contexts are

distinguished, which are listed below along with abbreviations in parentheses (see Table 1).

- (1) Faculty in a research university (FAC-PUB), where research universities are defined as those universities offering doctoral degrees in bioscience. Nearly all of these were rated in the Roose and Andersen (1970) study.
- (2) Faculty in nonresearch universities and four-year colleges, where publication is not likely to be a job requirement (FACNPB). These are primarily four-year colleges, and accordingly this context will sometimes be referred to as teaching in a college.
- (3) Research in locations where publication is assumed to be encouraged (RESPUB). These consist primarily of academic, private nonprofit, and governmental laboratories. In addition, this context includes those outstanding industrial laboratories in which publication is encouraged. This category does not include either prestigious academic research appointments or short-term academic research positions held prior to a tenure-track faculty appointment. Job titles include senior scientist, research associate, biochemist, scientist, and similar designations.
- (4) Research in locations where publication is not assumed to be encouraged (RESNPB), primarily industrial laboratories. For convenience this category will sometimes be referred to as industrial research.
- (5) Administration in any sector (ADMIN). During the period of the career that our

analyses will focus upon, this is primarily the administration of research in industrial laboratories.

- (6) Since the postdoctoral fellowship is an important career step for many biochemists (see below for further details), the category of postdoctoral fellow is included (FEL).

Several fundamental characteristics of the careers of biochemists are illustrated by the data in Table 1. First, the postdoctoral fellowship is the modal start of the career. Second, upon completion of the fellowship, a majority of fellows end up in faculty positions at research universities, followed by research positions in locations which encourage publication. Third, by the ninth year of the career there has been movement out of both types of research locations and an increase in the category of administration. In analyses not presented here, it is found that very little change among organizational contexts occurs, beyond what is necessary for the observed shift in the marginals. What change does occur is largely the increase in administration due to those who move from research to administrative roles.

THE ALLOCATION OF POSTDOCTORAL FELLOWSHIPS

Given the importance of the postdoctoral fellowship in terms of its numbers as noted above, its function in the training of biomedical scientists (cf. Curtis, 1969;

Table 1. Percentages of Scientists in Various Organizational Contexts over Time

Code	Category	% Immediately after Ph.D.	% 1st Postdoctoral, Postfellowship Location	% at Ninth Year of Career
FACPUB	Faculty in a research university	18.9	51.9	51.7
FACNPB	Teaching in a nonresearch university, primarily 4-year colleges	4.0	6.4	8.5
RESPUB	Research in locations where publication is encouraged	13.2	21.8	17.5
	% of category in academia	(34.7)	(43.7)	(33.7)
	% of category in government	(38.9)	(34.9)	(38.9)
RESNPB	Research in locations where publication is not encouraged (most industry and some government locations)	10.3	13.9	11.6
	% of category in industry	(100.0)	(97.4)	(92.1)
ADMIN	Administration in any location	4.4	5.9	10.3
FEL	Postdoctoral fellowships in any location, primarily research universities	49.2	0.0	0.4
		N = 545	N = 545	N = 543 ^a

^a Ns differ due to varying numbers of scientists excluded because of atypical work activities.

NAS, 1974), and its effects on later scientific achievement, the receipt of a postdoctoral fellowship is an important first step in many scientific careers. Research by McGinnis et al. (1981) has examined the allocation of fellowships along with the consequences of receiving a fellowship. The most significant result of their analyses is the lack of effect of predoctoral productivity on the odds of receiving a fellowship, whether productivity is measured by publications or citations. Thus, at least in biochemistry where postdoctoral fellowships are common, fellowships cannot be viewed as rewards for the demonstrated ability (to the extent that ability is reflected by published work) of young Ph.D.s. Rather, a combination of structural and personal effects dominate. Students from agricultural schools are much less likely to obtain fellowships. Young and unmarried students are more likely to obtain fellowships than older and married graduates. Finally, students from prestigious departments are more likely to receive a fellowship than those from less prestigious departments, independently of characteristics of the mentor.

ENTRANCE INTO THE FIRST JOB

The organizational context of a scientist's first job is of critical importance to his entire career. It provides both the physical resources with which to conduct research and the social context within which to work. Given the barriers to mobility which exist between the academic and nonacademic sectors (cf. Harmon, 1968), the first job is also important for its permanence.

To determine the extent to which various characteristics of a scientist determine the organizational context of his first job, the effects of three sets of variables are assessed. First, characteristics of the educational experience are considered, including the prestige of the doctoral department, the eminence and performance of the mentor, and the type of work experience had by the mentor. Second, the productivity of the student at the time he obtained the first position is considered, measured both with numbers of publications and citations. Finally, since the fellowship plays such an important role in

biochemistry, the fellowship itself is examined as a factor determining the choice among organizational contexts. This is done both by considering the simple fact of having a fellowship and by taking into account changes in productivity and prestige which result from the fellowship. Thus, if the fellowship was held in a graduate department, the prestige of that department replaces the prestige of the doctoral department in explaining the allocation of positions. And for those with fellowships, the level of productivity at the end of the fellowship replaces productivity at the time the doctorate was completed.

Our problem is to determine the effects of both continuous and discrete independent variables on a nominal dependent variable with five categories. This problem is ideally suited to multinomial logit analysis (cf. Domencich and McFadden, 1975; Nerlove and Press, 1973). Since sociologists are generally unfamiliar with this technique, a brief overview may be useful. Simply put, multinomial logit analysis estimates the effects of various independent variables on the odds of one outcome compared to another outcome. It is a generalization of log-linear models for dependent variables presented by Goodman (1971, 1972), allowing for continuous independent variables. When there are two outcomes, there is only one possible comparison. When there are more outcomes, such as in the current problem, a series of comparisons can be made. Thus, FACPUB can be compared to each of the categories FACNPB, RESNPB, RESPUB, and ADMIN. Similarly, FACNPB can be compared with RESNPB, RESPUB, and ADMIN, and so on, for a total of ten possible comparisons. The lack of ordering, let alone a metric, among the categories of the dependent variable thus adds considerable complexity to the analysis. Rather than being able to relate each independent variable to the dependent variable by means of a single coefficient, each independent variable requires a single coefficient for each of the nonredundant comparisons.²

² With k dependent categories there are $(k^2 - k)/2$ possible comparisons and thus possible coefficients for a given independent variable. However, only $k - 1$ of these will contain unique information.

The multiplicative logit coefficients labeled "b^s" in Table 2 indicate the factor by which the odds of one outcome versus another will change for a standard deviation change in a given variable, all other variables being held constant. Coefficients greater than one indicate an increase in the likelihood of the first outcome compared to the second outcome; coefficients less than one indicate a decrease in the relative likelihoods. For independent variables that are binary, such as having a fellowship, the unstandardized coefficient is also useful. It may be interpreted as the multiplicative change in the odds when an individual has the characteristic indicated by the dummy variable. For our purposes

one of the most useful outcomes of such analyses is the ability to rank outcome categories (i.e., FACPUB, RESPUB, RESNPB, FACNPB, and ADMIN) according to how strongly each independent variable differentiates alternative outcomes. An outcome that is ranked highly will become more likely for those with high values of the independent variable. The parameters described above are shown in Table 2; the rankings and parameter magnitudes are summarized in Figure 1.

Each of the subfigures in Figure 1 can be interpreted as follows. The horizontal axis represents the magnitude of the coefficients comparing a given outcome cate-

Table 2. Multinomial Logit Analysis of Determinants of First Postfellowship Job (N = 545)

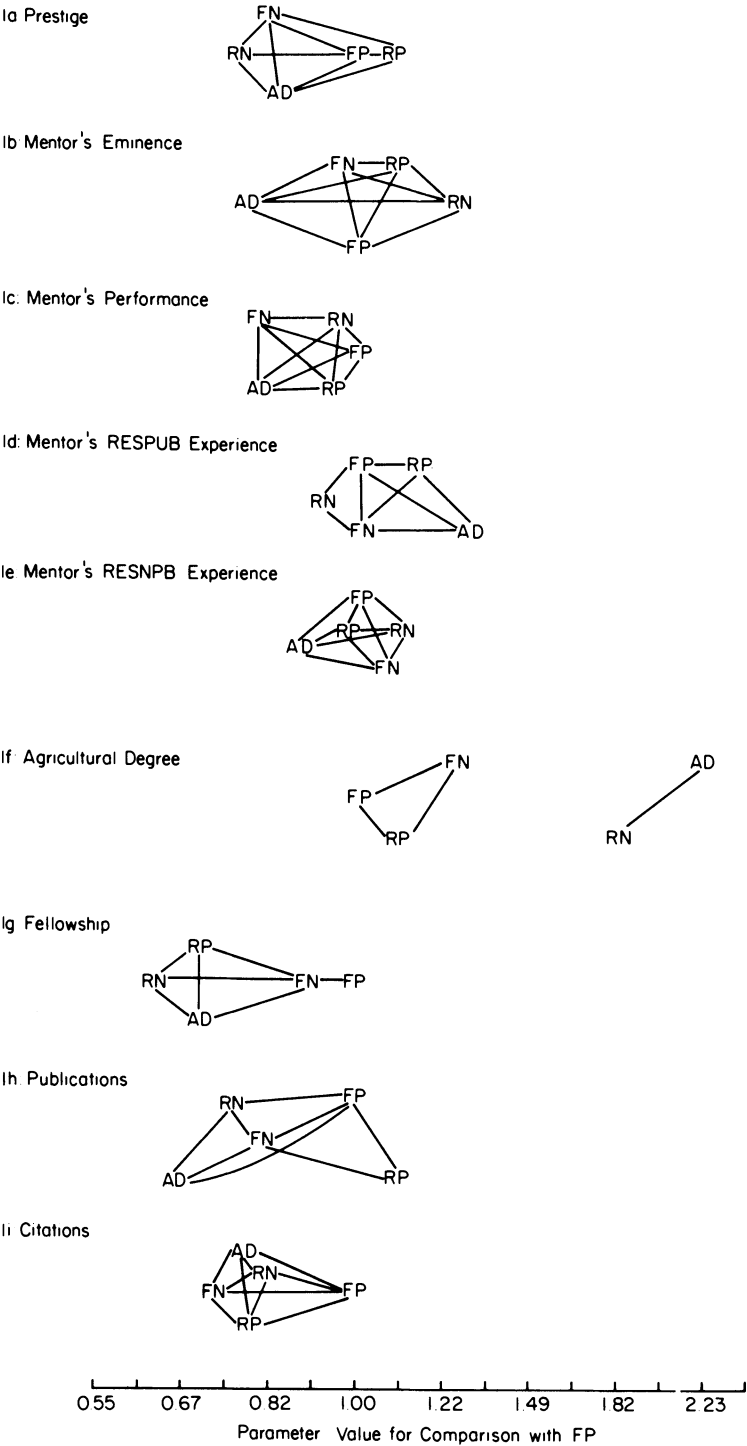
Comparison Groups		Coefficients of								
		PRST	MPRST	MPERF	MRESPUB	MRESNPB	AGDEG	FEL	PUB	CIT
1. RESPUB/ FACPUB	b ^s	1.05	1.10	0.93	1.14	0.99	1.09	0.70**	1.11	0.76
	b				1.31	0.99	1.20	0.50		
2. FACNPB/ FACPUB	b ^s	0.81	0.95	0.79	1.01	1.05	1.23	0.87	0.81	0.73
	b				1.01	1.16	1.53	0.76		
3. RESNPB/ FACPUB	b ^s	0.78	1.26	0.95	0.92	1.10	1.81**	0.64**	0.75	0.80
	b				0.84	1.29	3.48	0.40		
4. ADMIN/ FACPUB	b ^s	0.83	0.77	0.79	1.29	0.88	2.20**	0.70	0.66	0.77
	b				1.71	0.71	5.22	0.49		
5. FACNPB/ RESPUB	b ^s	0.77	0.87	0.85	0.89	1.06	1.12	1.24	0.73	0.95
	b				0.78	1.17	1.27	0.43		
6. RESNPB/ RESPUB	b ^s	0.74*	1.15	1.03	0.81	1.11	1.65**	0.90	0.68*	1.04
	b				0.64	1.31	2.88	0.82		
7. ADMIN/ RESPUB	b ^s	0.79	0.71	0.85	1.14	0.89	2.02**	1.00	0.60*	1.01
	b				1.31	0.72	4.33	1.00		
8. RESNPB/ FACNPB	b ^s	0.96	1.32	1.21	0.91	1.04	1.48*	0.73	0.93	1.09
	b				0.83	1.12	2.27	0.53		
9. ADMIN/ FACNPB	b ^s	1.02	0.81	1.00	1.28	0.83	1.80**	0.81	0.82	1.06
	b				1.69	0.62	3.40	0.65		
10. ADMIN/ RESNPB	b ^s	1.06	0.62	0.82	1.40	0.80	1.21	1.10	0.88	0.97
	b				2.04	0.55	1.50	1.22		

NOTE: Item identifications are: PRST = prestige of Ph.D. department if nonfellow, prestige of fellowship department if fellow; MPRST = prestige or eminence of mentor as weighted sum of awards (see text for details); MPERF = research performance or productivity of mentor during the three-year period after the student's Ph.D., measured as square root of citations to articles in that period; MRESPUB = dummy variable with value 1 if mentor has experience in a RESPUB context; MRESNPB = dummy variable with value 1 if mentor has experience in a RESNPB context; AGDEG = 1 if Ph.D. was from department in an agricultural school, 0 if not; FEL = dummy variable with a value of 1 if postdoctoral fellowship was held, 0 if not; PUB = square root of number of publications in the three-year period ending the first year of the job; CIT = square root of the number of standardized citations to publications in the three-year period ending the first year of the job.

Row b^s gives standardized logit coefficients indicating the change in the odds of the first context versus the second context for a standard deviation change in the given independent variable; row b gives unstandardized logit coefficients for dummy variables indicating the change in the odds for a unit change in the independent variable.

* Significant at .10 level, two-tailed test.

** Significant at .05 level.



KEY: FP=Faculty in a research university; FN=Faculty in a nonresearch university or college; RP=Research in a location encouraging publication; RN=Research in a location restricting publication; AD=administration. Contexts located to the right in figures are associated with higher values of the independent variable.

Figure 1. Summary of Multinomial Logit Analysis

gory to the baseline category of FACPUB, measured on a logarithmic scale.³ Thus, the further two categories are apart on the horizontal axis, the more strongly a standard deviation change in the independent variable differentiates them. If one category is to the right of a second category in the diagram, the odds of the first category are increased relative to the second category by an increase in the magnitude of the independent variable. All subfigures are drawn in the same scale and accordingly the relative distances between categories from one subfigure can be compared to those in any other subfigure. The distances on the vertical axis have no substantive interpretation, being added solely to facilitate the drawing of the figures. Finally, the lines between categories indicate the lack of a statistically significant difference (in terms of the given independent variable) between two categories. Thus, categories that are connected by a line are linked in the sense that the given independent variable does not significantly differentiate the likelihood of one outcome occurring compared to another. While general orderings were expected, these analyses are exploratory given the lack of prior research in this area. Accordingly, the test of significance represented by the lines are two-tailed tests at the .10 level of significance. Had one-tailed tests based on the suspected orderings been employed, somewhat stronger differentiation among categories would have been obtained. In evaluating the results presented in Figure 1, it is important to consider not only specific instances of statistical significance, but also the overall patterns of effects of a given variable.

Having a prestigious origin is associated with accepting a job in a location which encourages publication. Such locations,

which are generally considered to be the most prestigious, include faculty positions in research universities and research positions in Federally Funded Research and Development Laboratories, private non-profit laboratories, and academic research centers (FACPUB and RESPUB). An increase of one standard deviation in prestige, roughly the difference between a distinguished and a strong location on the Carter prestige scale, increases the odds of such positions over the contexts of administration, college teaching, and industrial research by an average factor of 1.27 (the inverse of the average of the effects in Equations 2 through 7 in Table 2). Thus, those from distinguished locations are 2.05 ($= 1.27^3$) times as likely to accept a position which encourages publication over any other type of job than their counterparts who obtained degrees in adequate-plus departments, all other variables being held constant. As shown graphically in Figure 1a, the prestige of the doctoral or fellowship department tends to differentiate organizational contexts into two groups. The first consists of faculty positions in research universities or research positions in locations which encourage publication; the second consists of contexts where some sort of structural constraint on publishing is often found.

Overall the effects of characteristics of the mentor on the organizational context of the student's first job are quite small. There are, however, several interesting configurations of effects. While prestige of the origin department and our measure of the mentor's eminence are significantly correlated ($r = .28$), the more eminent the mentor, the more likely a position where research is the primary role, regardless of encouragement or opportunity for publication. Thus, a prestigious departmental origin will decrease the chances of an industrial research position, while having an eminent mentor will increase the chances. Effects of the mentor's current research performance are not statistically significant, but nonetheless demonstrate a consistent pattern of differentiating research positions from nonresearch positions. Faculty positions in research universities and research positions in any sector are

³ The following relationship holds among additive parameters in a multinomial logit analysis, where d_{ij} is the additive logit coefficient comparing dependent categories i and j : $d_{ij} = d_{ik} - d_{jk}$. Hence, the relative distances among categories can be plotted in a one-dimensional space with no loss of information. Since we are using multiplicative parameters, which are simply the exponential transformations of the additive parameters, the coefficients must be plotted on a logarithmic scale.

positively associated with the mentor's research productivity.

The types of work experiences had by the mentor may affect the type of job obtained by the student. Accordingly, two dummy variables were coded indicating whether the mentor had work experience in either a RESPUB or RESNPB location. While the effects of these variables are not statistically significant, the most likely work activity of a student having a mentor with work experience in an industrial research position (RESNPB) is an industrial research position. The most common organizational context of those having mentors with nonfaculty, academic research experience (RESPUB) is either administration (which could be in a lab encouraging publication) or research in a location encouraging publication.

Having an agricultural degree is the strongest factor affecting choice of context, even after controlling for its effect on obtaining a fellowship. Those with degrees from agricultural schools are on average nearly five times as likely to accept administrative positions and over three times as likely to accept industrial research positions than positions encouraging publication (Equations 3, 4, 6, and 7). Those from agricultural schools are also significantly less likely to accept teaching positions in nonresearch universities and colleges (Equations 8 and 9). As shown in Figure 1f, whether or not a degree is obtained in an agricultural school divides organizational contexts into two distinct groups. The first group, which is associated with an agricultural degree, consists of primarily industrial positions. The second group consists of nonindustrial contexts, with teaching in a research university or doing research in a location encouraging publication being least associated with an agricultural degree. While it is impossible to determine the exact process leading to these outcomes, they suggest both a greater growth in nonagricultural biochemistry in the universities, resulting in more teaching positions for nonagricultural biochemists than agricultural biochemists, and a greater demand for agricultural biochemists in the food industry.

Postdoctoral fellows are nearly twice as

likely to obtain faculty positions in research universities (FACPUB) as jobs in any other context (Equations 1 to 4), all other factors being held constant. The fellowship does not significantly differentiate among the remaining contexts, although college teaching is more likely than any of the nonfaculty contexts that restrict publication. Overall, postdoctoral fellowships lead to academic positions, primarily in research universities.

Citations at the time the first job is obtained have no effects on the likelihood of various job alternatives (see Figure 1i). The number of citations to a student's pre-employment publications increases the odds of a faculty position in a research university by a factor of 1.3, although this tendency is not statistically significant.

Number of publications has a stronger effect, differentiating organizational contexts which encourage or demand publication from those which explicitly or implicitly restrict publication. Figure 1h shows that industrial research jobs and administrative positions are associated with few publications, jobs in research universities and locations encouraging publication (RESPUB and FACPUB) are associated with more publications, and college teaching falls in between the two other groups.

The fact that citations have no effects at all in the allocation of positions, while number of publications does differentiate those positions which strongly encourage publication from those which discourage publication, suggests that scientists who have published and want to continue publishing seek out and obtain positions where they will be rewarded for an activity at which they have already been successful, as opposed to positions where publication is simply tolerated. Past studies which have reported that the "best" students go to nonindustrial locations may have equated "best" with "most publications," regardless of the quality of those publications.

Overall our findings suggest several things. Productivity is important only in terms of quantity of publications. Those who publish before obtaining a permanent job have a tendency to obtain jobs which encourage publication. The nature of the

graduate education and the postdoctoral fellowship have a variety of effects. Prestige is associated with accepting a faculty position in a research university and with not obtaining an industrial research position. Having an eminent mentor, on the other hand, is associated with accepting a research position as opposed to a faculty position. Beginning the career with a postdoctoral fellowship increases the chances of obtaining a faculty position in a research university.

In assessing these findings it is important to realize that these are partial effects which remain after statistically controlling for all other variables in the logit analysis. Thus, since fellows tend to increase their origin prestige by replacing the prestige of the doctoral department by that of the postdoctoral location and to increase their level of productivity during the period of the fellowship, and since both of these changes will increase the odds of a faculty position in a research university over an industrial research position, the total effect of the fellowship is greater than the direct effect of the dummy fellowship variable. Further, while a nonagricultural degree directly increases the likelihood of a faculty position in a research university, it also increases the likelihood through its effect on the acceptance of a postdoctoral fellowship.

THE EFFECTS OF ORGANIZATIONAL CONTEXT ON PRODUCTIVITY

To control for the effects of changes in organizational context on productivity, our analysis is initially restricted to those scientists who remain in the same job for at least nine years. This period of time should be long enough to allow the effects of organizational context to emerge. To assess the effects of changes in organizational contexts and to determine how responsive a scientist's productivity is to a change in his work environment, those scientists who change contexts are then examined.

The effects of organizational context are introduced into regressions on productivity by means of a set of four dummy variables, one coefficient for each of the categories of administration (ADMIN),

research in a location encouraging publication (RESPUB), research in a location limiting publication (RESNPB), and teaching in a college (FACNPB). Having a faculty position in a research university (FACPUB) is the fifth, excluded category that serves as a standard of comparison for the other contexts.

After three years in the job, significant effects of context on level of publication begin to appear (Equation 1, Table 3). Having a teaching position in a college (FACNPB) decreases the level of publication relative to faculty in research universities by an average of .56 on the transformed scale of publications, after controlling for characteristics of the doctoral and/or postdoctoral experience.⁴ The effect of an industrial research position (RESNPB) is a relative decrease of .74 publications. There are no significant differences in publication levels between faculty in research universities and the two remaining categories.

To take into account the initial differences in productivity among scientists entering different organizational contexts, Equation 2 controls for the productivity level of a scientist at the time he begins his job. In such an equation, the effect of the contextual variables can be thought of as the change in productivity that has occurred after a scientist has entered a particular organizational context. In short, the effects of differing levels of productivity upon entering the job have been removed. The strongest effect is now stability. A scientist who publishes before entering a particular context tends to continue publishing, independently of context. Still, organizational context remains a significant factor, albeit with attenuated effects. Working in industrial research (RESNPB) has the strongest depressing contextual effect with an unstandardized coefficient of $-.49$; the second largest effect is for college teaching with a coeffi-

⁴ A square root transformation has been made for both publications and citations; thus the differences between adjacent productivity counts (e.g., five and six publications; nine and ten citations) decrease as the level of productivity increases. In this particular case the use of the transformation implies that contextual effects are stronger in terms of raw publications among the more productive scientists in a location.

Table 3. Determinants of Publications for Those Staying at the Same Job for at Least Nine Years (N = 308)

Publication Year		Coefficients of										R ²
		PRST	MPRST	MPERF	AGDEG	FEL	RESPUB	FACNPB	RESNPB	ADMIN	(From Year One) PUB CIT	
1. Year 3	b	.0343	-.00488	.0390*	-.136	.273**	.0620	-.559**	-.741**	-.141		.143
	b ^s	.035	-.021	.101	-.066	.136	.024	-.159	-.243	-.041		
2. Year 3	b	.0407	.00187	-.0113	-.0272	-.00328	.132	-.346**	-.493**	-.0157	.615**	.460
	b ^s	.041	.008	-.029	-.013	-.002	.051	-.098	-.162	-.004	.560	.074
3. Year 6	r	.116	.049	.155	-.146	.193	.063	-.149	-.253	-.045	.646	.498
	b	.104*	.000250	.0469**	-.0498	.283**	-.0233	-.961**	-.936**	-.473**		.193
	b ^s	.090	.001	.104	-.021	.121	-.008	-.234	-.263	-.117		
4. Year 6	b	.109**	.00558	.00284	.0533	.600	.0419**	-.802**	-.753**	-.408**	.391**	.325
	b ^s	.094	.021	.006	.022	.026	.014	-.195	-.212	-.101	.305	.125
5. Year 9	r	.199	.093	.210	-.108	.200	.053	-.220	-.239	-.104	.472	.410
	b	.165**	-.00782	.0525**	-.103	.283**	.0508	-.885**	-.961**	-.391**		.206
	b ^s	.139	-.208	.114	-.042	.118	.016	-.210	-.264	-.095		
6. Year 9	b	.172**	-.00199	-.000956	.0310	.0319	.134	-.722**	-.778**	-.360**	.316**	.350
	b ^s	.145	-.007	-.002	.013	.013	.043	-.172	-.214	-.087	.241	.221
	r	.236	.077	.225	-.118	.201	.071	-.209	-.253	-.093	.473	.455
7. Year 9	b	.0779*	-.00834	.0182	-.0341	.0276	.0792	-.240*	-.367**	-.0671	(From Year Six) PUB CIT	.592
	b ^s	.066	-.030	.040	-.014	.012	.026	-.056	-.101	-.016	.582**	.0860**
	r	.236	.077	.225	-.118	.201	.071	-.209	-.253	-.093	.569	.164
											.747	.612

NOTE: Item identifications are: RESPUB = 1 if organizational context is RESPUB, 0 if not; FACNPB = 1 if context is FACNPB, 0 if not; RESNPB = 1 if context is RESNPB, 0 if not; ADMIN = 1 if context is ADMIN, 0 if not; FACPUB is the excluded context; see Table 2 for other item identifications.

Row b gives unstandardized regression coefficients; row b^s gives standardized regression coefficients; row r gives zero order correlations with dependent variable.

* Significant at .10 level, one-tailed test.

** Significant at .05 level.

cient of $-.35$. The effect of administration remains small and insignificant; the effect of research in a location that encourages publication is positive (although not significant at the $.10$ level) and of increased magnitude, suggesting a slight increase in the productivity levels of these scientists.

The effects of characteristics of the doctoral and postdoctoral experience change significantly when change in the level of publication is examined (Equation 2), rather than the level of publication (Equation 1). While having a fellowship significantly increases the *level* of publication, adding $.27$ transformed publications, it has no effect on *changes* in the level of publication which occur after the job is obtained. Thus, the experience of the fellowship does not directly affect subsequent levels of productivity. Rather, its effect is indirect, operating via its positive effect on productivity during the period of the fellowship and its effect on the allocation of jobs. Other variables related to the doctorate and fellowship have no significant effects on either level of or change in publication, with the exception of the performance of the mentor, which has a barely significant positive effect. In regressions on level of publication excluding contextual variables (regressions not shown), the effects of all of the independent variables were stronger with increased levels of statistical significance. It appears that most of the effects of these characteristics occur in the form of cumulative advantage. They affect early productivity and initial job placement, which in turn affect later productivity, independently of their own determinants.

The effects of educational characteristics remain basically unchanged at the sixth year (Equation 3), with an increase in the magnitude and significance of the effect of origin prestige on both level and change in publications, and an increased effect of the mentor's performance on level of publication. The effects of context on level of publication in the three-year period ending in year six of the job are similar to those found in year three, with the exception of the expected emergence of the negative effect of administration on the level of publication.

The prestige of the origin has a positive

effect on the change in publication level over the first six years of the job. Simply having a fellowship, attending an agricultural school, or having an eminent or productive mentor do not. The strength of the contextual effects has increased. Compared to those with faculty positions in research universities, those in administration have significantly decreased their rate of publication by $.51$ transformed articles over a three-year period. Such a decrease is even stronger for those in colleges and industrial research, with decreases of nearly one article. There are no significant differences between faculty in research universities and researchers in locations encouraging publication. Finally, as before, there is a strong effect of the past level of publication on the present level of publication, although the magnitude has decreased slightly, as would be expected due to the greater lag between productivity measures. Past citations are beginning to have a positive effect on changes in productivity, suggesting the emergence of a reinforcing effect of recognition on publication.

By the ninth year the effects on level of publication of having a prestigious origin or a productive mentor have increased, while the effect of having a fellowship is unchanged (Equation 5). The effects of context on the level of publication are essentially unchanged from year six. Equation 6 continues the trend of effects on changes in level of productivity that appeared in Equation 4.

To assess the relative strengths of the effects of earlier productivity and organizational context, a sheaf coefficient (Heise, 1972) for the contextual variables can be computed. Its value for Equation 6 is 0.28 , indicating a stronger overall effect of context than either number of publications or number of citations, and a slightly weaker effect than the combined effects of early citations and publications. Productivity at the start of the job remains a strong factor predicting future productivity, although its effect decreases with time, while the effect of context increases.

The nature of the effect of organizational context on productivity is further illustrated in Equation 7 of Table 3 in which level of publication in year 9 is re-

gressed on the same variables as in Equation 6, with the substitution of levels of productivity from year six for productivity in year one. In this equation the effects of the contextual variables on changes in level of publication are of only slightly decreased magnitudes. Thus, organizational context has a continuing effect on level of publication lasting at least nine years into the job.

Similar analyses are possible with the number of citations received by scientists in various contexts, where citations can be thought of as an indicator of the recognition that published work has received (see Table 4). Results are similar to those obtained for publications.

As with publications, the productivity measures in Table 4 are for the three-year period ending in a given year of the job. How many years this is from the year of the doctorate depends on whether or not a postdoctoral fellowship was held before the first job. Thus, the citation measure for year six is computed from citations to articles in the fourth through sixth years of the job. Citations were coded from a volume of *Science Citation Index* published one or two years after the sixth year. Complete details on the construction and standardization of these measures are presented in Long (1978).

While the contextual effects are similar, although taking somewhat longer to emerge, the effects of background characteristics differ. For all years examined (see Equations 1, 3, and 5), an agricultural degree significantly decreases the number of citations received, a prestigious origin increasingly increases citations, and a productive mentor increases citations. While having a fellowship has a strong positive effect on citations in the third year, the effect diminishes by the sixth year and turns negative by year nine. Thus, even though the prestige of the fellowship along with the prestige of the doctorate continue to positively affect the level of productivity, simply having a fellowship does not maintain its positive effect after controlling for other factors.

By the third year in a given context, there is one significant contextual effect on the level of citations—the negative effect of more than one transformed citation

for those who hold industrial research positions compared to faculty in research universities. By year nine there are also significant negative effects for faculty in colleges and administrators. Finally, those with research positions in nonindustrial locations (RESPUB) are not significantly differentiated from faculty in research universities.

Equations 2, 4, and 6 examine the effects of educational characteristics and contextual effects on changes in levels of citation. In each equation productivity at the first year of the job is controlled. By year six negative and statistically significant effects of administration, college teaching, and industrial research are found; by year nine these effects have increased, with a sheaf coefficient of 0.20. Hence, not only are levels of citation negatively affected by context, but so are the changes in citations received. As with publications, it takes time before the contextual effects emerge. Equation 7 shows that the negative effects continue, affecting the change in citations from the sixth to the ninth year in the job.

When changes in level of citation are examined, effects of educational variables change. Having an agricultural degree no longer has a significant effect, nor does the eminence of the mentor. Having a prestigious fellowship or doctoral department, however, strongly increases the change in level of productivity. Simply having a fellowship has an initial positive effect which becomes negative by year nine.

Organizational context emerges as a strong factor determining not only levels of productivity, but also, and more importantly, changes in rates of productivity occurring after a position is obtained. Administrators, faculty in colleges, and researchers in locations limiting publication decrease their productivity relative to faculty and researchers in research universities. These contextual effects go beyond initial differences in doctoral education and pre-employment productivity, which were shown to affect the allocation of positions. For after controlling for these variables, contextual effects remain strong and statistically significant. Accordingly, it is not that those in research

Table 4. Determinants of Citations for Those Staying at the Same Job for at Least Nine Years (N = 308)

Citation Year	Coefficients of											R ²
	PRST	MPRST	MPERF	AGDEG	FEL	RESPUB	FACNPB	RESNPB	ADMIN	(From Year One) PUB CIT		
1. Year 3	b	.191*	-.0124	.129**	-.666**	1.33**	.266	-.307	-1.05**	.197		.245
	b ^s	.090	-.025	.156	-.151	.311	.048	-.041	-.161	.027		
2. Year 3	b	.210**	.00166	-.0331	-.208	.687**	.546**	.014	-.713**	.00933	.0387	.642**
	b ^s	.099	.003	-.040	-.047	.160	.098	.002	-.109	.001	.016	.613
	r	.197	.103	.232	-.240	.380	.015	-.066	-.229	-.049	.522	.690
3. Year 6	b	.309**	.00437	.0811*	-.463**	1.05**	-.172	-.988**	-.566*	-.564*		.177
	b ^s	.137	.008	.092	-.099	.231	-.029	-.123	-.082	-.072		
4. Year 6	b	.323**	.0158	-.0292	-.179	.554**	.00432	-.679**	-.223	-.545*	.508**	.306**
	b ^s	.143	.030	-.033	-.038	.121	.001	-.085	-.032	.203	.275	
	r	.238	.132	.220	-.163	.312	-.037	-.138	-.096	-.116	.470	.496
5. Year 9	b	.408**	.00932	.115**	-.695**	-.814**	-.132	-.128**	-1.21**	-.761**		.159
	b ^s	.176	.017	.127	-.144	.174	-.022	-.155	-.170	-.094		
6. Year 9	b	.420**	.0185	.0184	-.434*	-1.22**	.0282	-1.05**	-.966**	-.810**	.228	.326**
	b ^s	.181	.034	.020	-.090	.261	.005	-.127	-.136	-.100	.088	.285
	r	.220	.105	.239	-.142	-.059	.051	-.167	-.144	-.088	.306	.356
7. Year 9	b	.241**	.00739	.0638*	-.482**	-1.36**	-.0514	-.514*	-.637**	-.352	(From Year Six) PUB CIT	.405
	b ^s	.104	.014	.071	-.100	-.290	-.008	-.062	-.089	-.043	.363**	.419**
	r	.220	.105	.239	-.142	-.059	.051	-.167	-.144	-.088	.181	.408
											.500	.527

NOTE: See Tables 2 and 3 for item identifications.

Row b gives unstandardized regression coefficients; row b^s gives standardized regression coefficients; row r gives zero order correlations with dependent variable.

* Significant at .10 level, one-tailed test.

** Significant at .05 level.

universities are more capable, as indicated by previous levels of publication and citation, or better trained, as indicated by characteristics of their graduate education, than their counterparts in other contexts. The fact that it takes time for the contextual effects to emerge suggests that these effects are not simply the result of immediate changes in scientists' goals or categorical barriers to publication in some contexts. Scientists in administration, industry, and colleges begin their employment in those locations with continued levels of productivity, but in time the effects of earlier productivity decrease while the effects of context increase.

PRODUCTIVITY, ORGANIZATIONAL CONTEXT, AND MOBILITY

Changes in organizational context are relatively infrequent. For the 557 scientists in this study, only 275 changes in context (as defined here) involving 208 individuals occurred. Thus, for the 6915 career-years for which data are available, changes occur on an average of once every 25 years, affecting 37% of the sample. A majority of these changes are in roles rather than sectors, such as moving into academic administration from a research or faculty position in academia, or moving from research to administration in industry.

Of the 275 moves, 69 were selected for analysis. To ensure independence of observations in the regression analyses, multiple moves by the same individual were excluded, reducing the number of possible moves to 208. Of the remaining moves, 44% were excluded because of censoring on the second job before it lasted six years; in such cases no information was available regarding the scientists' eventual duration in the second job. To allow for comparability with the analyses of the stayers, 47 moves were excluded that were not censored, but which had a first job lasting less than three years or a second job lasting less than six years. Of these, 47% involved jobs lasting one year. The remaining sample included 69 moves. Analyses were also run with the 94 job pairs that had at least two years of data for each job. Results were similar, although

as would be expected by the limited amount of time for contextual effects to operate, the magnitudes of the effects were attenuated. Given the small size of the sample, the results must be viewed with consequent caution.

Equation 1 of Table 5 regresses publications at the time of the job change on educational characteristics, the *first* organizational context, and measures of publications and citations three years prior to the move.⁵ Except for the barely significant, negative effect of the mentor's eminence, the effects of educational characteristics are not significant. The effects of contextual locations are in the expected directions and are significant at the .10 or higher level. As for the stayers, the effect of research in an industrial location is large, as is the effect for teaching in a college. Unlike the case for stayers, there are significant differences between faculty at research universities and researchers in such locations. Finally, the effect of past levels of productivity on current levels of publication are positive. The effect of publications is strongest and significant at the .01 level; the effect of citations is significant and approximately two-thirds the size of the publication effect.

Equation 2 presents comparable results for the sixth year in the second organizational context. The independent variables now include indicators of the *second* organizational context and levels of productivity in the *three-year period before* the change of contexts. The similarities between Equation 2 of Table 5 and Equation 4 of Table 3 for stayers are important, as is the one major difference. Educational characteristics do not have significant effects (which in part reflects the small sample size). Organizational contexts have effects in the same directions and of roughly similar magnitudes. Interestingly, the effects of past productivity on current productivity are greatly at-

⁵ Three-year publication counts for the dependent variable are for the period ending the first year in the second job. This lag of one year is introduced to allow for the time between the submission of manuscripts and their publication. Because these scientists had their first job for an average of 5.1 years, the results would be expected to fall between those presented in Equations 2 and 4 of Table 3 and 4.

Table 5. Determinants of Publications for Movers Staying in the Second Job at Least Six Years (N = 69)

Publication Year	Coefficients of										R ²
	PRST	MPRST	MPERF	AGDEG	FEL	RESPUB	(Context of First Job)		ADMIN	(Three-Year Period Ending Two Years Before Move)	
1. Year 1 in 2nd Context	b	.015	-.0712*	.0108	-.226	-.321	-.560**	-1.32**	-1.14**	.406**	.114**
	b ^s	.014	-.204	.023	-.110	-.137	-.266	-.299	-.512	.384	.265
	r	.070	-.150	.150	-.226	.077	.216	-.007	-.431	.582	.487
2. Year 6 in 2nd Context	b	.120	.0291	-.0278	-.0943	.244	.180	-1.33**	-.567*	.158	.00925
	b ^s	.118	.087	-.065	-.048	.109	.059	-.278	-.185	.165	.020
	r	.213	.128	.127	-.170	.256	.207	-.215	-.140	.223	.295

NOTE: See Tables 2 and 3 for item identifications.
Row b gives unstandardized regression coefficients; row b^s gives standardized regression coefficients; row r gives zero order correlations with dependent variable.
* Significant at .10 level, one-tailed test.
** Significant at .05 level.

tenuated. That is, the productivity that resulted from work in the first organizational context has a much smaller effect on level of publication in the current context. This differs both from the results in Equation 1 of Table 5 and Equation 4 of Table 3 for the stayers. Within the same context, past productivity influences current productivity. Once mobility occurs across sectors, only organizational context significantly affects publication levels. Past organizational context of employment has no further effect on productivity in the new context (regression not shown). This lack of effect further underscores the potency of current organizational context as a determinant of productivity.

Equation 1 of Table 6 shows that educational characteristics have no significant effects on changes in the citation level within the first context. Moderate effects of organizational context are found in the expected directions, but are generally significant at only about the .20 level. Industrial research is the one exception, being significant at the .05 level. Finally, past levels of citation influence current levels, with the standardized coefficient being large ($b^s = .56$) and significant at the .01 level.

In Equation 2 the results are presented for the sixth year in the *second* context. As with publications, productivity during the last three years in the *first* context has a smaller effect than for stayers. Effects of contextual location are in the expected direction. As with publications, the effects of the first context are very small and not nearly significant (regression not shown). One interesting difference is that educational characteristics emerge as significant factors affecting changes in citation level. Having an eminent mentor positively affects changes in citation level at the new job. Obtaining a degree from an agricultural school has a strong negative effect.

The major finding of this section is that when changes of organizational context occur, changes in productivity follow. These changes are not strongly affected by past productivity or past organizational context. Rather, the organizational context of the new position quickly emerges as the significant factor influencing pro-

ductivity. Since contextual effects on levels of productivity in the movers' first context conform to the results obtained for the stayers, it is unlikely that the movers represent those who were unsuccessful in meeting the expectations of their first job context. The argument that the most productive scientists end up in the most prestigious locations is not supported by our data, since, when mobility occurs, the once-productive scientists suffer declines in their productivity—declines that are determined by the new organizational context. The argument would hold only if one were willing to assume that a scientist's ability to make contributions to the body of scientific knowledge changes along with a change of organizational context. This, however, seems unlikely.

SUMMARY AND CONCLUSIONS

In an earlier study of the academic career, Long (1978) examined the relationship between a scientist's productivity and the prestige of his academic position. He found that while a scientist's productivity does not affect the prestige of the academic position obtained, the prestige of the position does affect the scientist's later productivity. Those in prestigious locations increased their current productivity independently of earlier productivity. Our current study shows that scientific productivity and organizational context are similarly related. The chances of obtaining employment in a particular organizational context are not strongly affected by productivity as indicated by either published papers or the scientific community's utilization of those papers. While there is some tendency for those who have published to obtain positions in contexts which encourage publication, no similar effect is found when citations are considered. Within three to six years of obtaining a position in a specific context, a scientist's level of productivity conforms with the characteristics of that context, independently of previous productivity. Compared to those in research universities, those doing industrial research, teaching in four-year colleges, or administering a laboratory decline in the

Table 6. Determinants of Citations for Movers Staying in the Second Job at Least Six Years (N = 69)

Citation Year	Coefficients of										R ²
	PRST	MPRST	MPERF	AGDEG	FEL	RESPUB	FACNPB	RESNPB	ADMIN	(Three-Year Period Ending Two Years Before Move) PUB CIT	
1. Year 1 in 2nd Context	b	.181	-.0501	-.0167	.183	.259	-.498	-1.33	-1.20**	.192	.488**
	b ^s	.084	-.071	-.018	.044	.055	-.117	-.149	-.265	.090	.563
	r	.212	.058	.228	-.305	.327	.263	-.074	.344	.524	.672
2. Year 6 in 2nd Context	b	.140	.215*	.0391	-1.55**	.509	-.626	-2.63**	-1.02	-1.20**	.420
	b ^s	.051	.237	.033	-.292	.084	-.075	-.201	-.112	-.220	.265
	r	.272	.343	.276	-.397	.300	.070	-.161	-.151	-.228	.403

NOTE: See Tables 2 and 3 for item identifications.
Row b gives unstandardized regression coefficients; row b^s gives standardized regression coefficients; row r gives zero order correlations with dependent variable.
* Significant at .10 level, one-tailed test.
** Significant at .05 level.

number of and citations to their publications. The importance of context is further demonstrated by the flexibility in productivity patterns which is observed among those who move from one context of work to another. Within six years the levels of productivity of the movers have changed to the extent that past productivity no longer predicts future productivity. New levels of productivity are determined largely by the context of work.

These results do not support the idea that scientists are allocated to various contexts on the basis of contributions to the body of scientific knowledge. Past research indicating that the most productive scientists are recruited to academic locations may have confused the *cause* of a scientist working in a given organizational context with the *effect* of working in that context.

Our findings provide additional evidence for the operation of cumulative advantage in the scientific career (cf. Allison and Stewart, 1974; Cole and Cole, 1973): initial advantages are soon transformed into additional advantages. For example, having a prestigious doctoral department increases the chances of the graduate obtaining a postdoctoral fellowship or a position in a research university. The position in a research university results in increased productivity; the fellowship directly increases publications over the fellowship period and indirectly increases productivity by increasing the chances of obtaining a faculty position in a research university, thus providing a secondary effect of doctoral prestige. The fact that the major determinants of productivity a decade into the career are not direct effects of education and training, but largely the effects of past productivity and organizational context of employment, points to the extreme importance of the recruitment process in explaining inequalities among scientists in productivity. That recruitment operates independently of productivity suggests that, to the extent that publications and citations reflect a scientist's contribution to scientific knowledge, the stratification system in science is far from universalistic. Rather, initial advantages are accumulated, resulting in the well-known inequality among scientists in levels of productivity.

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